Title 1:

Efficient Prediction of Accuracy in Inflation Rate Using Extreme Gradient Boosting Algorithm in Comparison with Decision Tree Algorithm

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keywords: Prediction, Extreme Gradient Boosting, Decision Tree, Inflation Rate, Accuracy, Dataset.

Abstract:

Aim: The purpose of this study is to assess the effectiveness of various machine learning algorithms in order to increase the forecast accuracy of inflation rates. In particular, it evaluates the effectiveness of the Decision Tree approach and the XGBoost (extreme gradient boost) algorithm to see which produces the most accurate predictions. **Materials and Methods:** A dataset of 15 years' worth of previous rates of inflation records for 19 different nations was used. Financial metrics, modifications to policy, and other pertinent characteristics are included in the dataset. 10 iterations of each algorithm were conducted, with 10 participants per group. The same dataset was used to train both models with the same set of parameters. The resulting true-positive (TP) rate was 0.132. **Result:** The investigation shows that the two approaches' accuracy in forecasting differs significantly. With a precision of 90.11%, the XGBoost (extreme gradient boost) model beat the Decision Tree approach, which had an efficiency of 77.25%. **Conclusion:** According to the study, the XGBoost algorithm predicts inflation rates more accurately than the Decision Tree approach. When it comes to accurately predicting inflation rates, XGBoost outperforms the Decision Tree model with a precision of 90.11% as opposed to 77.25%. As a result, it is suggested as the best method for accurately predicting inflation rates.

Keywords: Decision Tree, Comparison, Extreme Gradient Boosting, Prediction, Accuracy, Inflation Rate.

Introduction:

Financial planning, investment plans, and economic policies are all greatly influenced by accurate inflation rate forecasting. Because machine learning algorithms can examine intricate patterns in vast datasets, they have become indispensable tools for inflation predictions. XGBoost (extreme gradient boosting) and decision tree methods are two of these methods that provide unique perspectives on predictive modeling. Decision trees offer a straightforward, understandable framework for prediction, while XGBoost improves accuracy by continually refining weak models using boosting methods. The purpose of this study is to evaluate how well its XGBoost and decisions tree algorithms estimate inflation rates in order to identify which approach offers the most accuracy and dependability(Vilenko 2025).

This study's main goal is to assess both systems' prediction capabilities using historical inflation data. Decision trees are popular because they can separate data according to feature relevance and produce simple predictions. However, their generalizability is limited by their propensity to overfit. By lowering bias and variance, XGBoost, on the other hand, maximizes the learning process and is hence more suited for intricate economic datasets. This study aims to determine which model produces the most accurate and efficient inflation rate projections by doing a comparison analysis(Tsuchiya, Takemura, and Udagawa 2025).

Managing the dynamic character of inflation, which is impacted by several regional and global causes, is one of the main issues in economic forecasting. A model's capacity to adjust to these fluctuations and derive significant insights determines its predictive accuracy. By examining the advantages and disadvantages of the XGBoost & Decision Tree algorithms, this study fills a research gap. This study is to identify the most effective strategy for enhancing inflation rate forecasts by using actual economic data, which would eventually improve financial decision-making and promote economic stability(Beer, Ferstl, and Graf 2025).

Materials And Methods:

The purpose of this study was to assess how well various machine learning models predicted inflation rates inside SIMATS. The models were divided into two groups by the study: the first group used the Decision Tree method, while the second group used the XGBoost (extreme gradient booster) technique. Alpha = 0.05 whereas beta = 0.2 were used for calculations, and both models were run at different intervals.

Inflation rate records from several nations over a 19-year span, including economic metrics and changes in policy, make up the dataset utilized in this study. Frequent pre-processing improves the quality of the dataset and results in more accurate analysis. To guarantee accuracy and consistency, a variety of data cleansing strategies were used.

Extreme Gradient Boosting:

Iterative boosting is used to train the XGBoost (extreme gradient boosting) algorithm. To reduce mistakes and improve performance, XGBoost incorporates weighted updates and regularization strategies during model generation. Predictions become more accurate as a result of this boosting mechanism's reduction of bias and variation. Superior handling of intricate interactions, effective computing, regularization-based resistance to overfitting, and adaptation to big datasets are just a few benefits of XGBoost. Better model interpretability is also facilitated by

its efficient feature significance determination. All things considered, XGBoost is well-liked for its effectiveness, scalability, and remarkable predicted accuracy on a variety of workloads.

Decision Tree:

A centralized, branch-like design that divides data according to feature requirements is used to train the Decision Tree algorithm. To reduce impurity and increase the accuracy of classification or regression, the model builds by choosing the most important features at each node. Decision-making processes may be easily interpreted and visualized with the aid of this organized technique. Among the many benefits of decision trees are their ease of use, quick calculation, little need for data pretreatment, and capacity to handle information that is numerical as well as categorical. Nevertheless, pruning approaches can help reduce their propensity for overfitting. All things considered, decision trees are prized for their openness, usability, and efficiency in resolving a range of prediction problems.

Statistical Analysis:

IBM SPSS is used for quantitative analysis, which assesses the precision of inflation rate models for forecasting. To determine the importance of variations in accuracy values, methods like testing hypotheses and the cross-validation are used. The aim of this research is to conduct a thorough quantitative analysis and identify the best method for raising the predicting accuracy of inflation rates.

Result:

When it came to forecasting inflation rates, the XGBoost (extreme gradient booster) algorithm outperformed the Decision Tree method in terms of accuracy. The accuracy of the Decision Tree method was 77.25%, whereas XGBoost's precision was 90.11%. In every iteration, XGBoost consistently performed better than the Decision Tree model, as seen in Table

A statistical assessment of the two approaches is shown in Table 4. The stability and robustness of the XGBoost model were demonstrated by its mean reliability of 85.75% without an acceptable standard variation (SD) of 2.914. On the other hand, the Decision Tree method showed poorer prediction dependability with a mean accuracy equal 74.30% and an average variance (SD) of 3.128. These results demonstrate that X G Boost is the superior method for precisely predicting inflation rates.

Discussion:

According to this study, the XGBoost (extreme gradient booster) method performs better than the Decision Tree approach in the field of price projections, with an accuracy rate = 90.11% as opposed to 77.25%. The advantage of XGBoost in capturing intricate correlations in inflation data was further supported by a paired t-test that verified a statistically significant variance amongst the two models. The improved accuracy of XGBoost shows how well it can handle economic patterns, cut down on mistakes, and produce forecasts that are more trustworthy. These results demonstrate how well XGBoost predicts inflation, which makes it a useful tool for financial analysts and policymakers(Ray et al. 2025).

These outcomes are corroborated by related research, which shows that boosting-based models, such as XGBoost, routinely perform better in predicting tasks than conventional decision tree-based techniques. Nonetheless, other academics contend that decision tree models are helpful in situations where transparency is important since they provide faster computation times and improved interpretability despite their lower accuracy. However, most people agree that XGBoost has a major advantage over decision tree algorithms for finance modeling scenarios because of its capacity to reduce bias & variation(Huang, Qi, and Xia 2024).

This study supports the widely held belief in the machine learning and financial worlds that XGBoost offers better accuracy for predicting inflation rates. These findings' agreement with earlier studies confirms XGBoost's dependability in identifying inflation patterns. Future research should examine how feature selection, hyperparameter tweaking, and other economic indicators affect prediction accuracy. Gaining a better grasp of these elements can aid in forecasting model optimization, producing more accurate and useful information for financial planning and decision-making(“Global Inflation Forecasting and Uncertainty Assessment: Comparing ARIMA with Advanced Machine Learning” 2025).

Conclusion:

According to this study, the XGBoost (extreme gradient booster) algorithm performs noticeably better than the Decision Tree method in terms of more accurate inflation rate prediction. The Decision Tree algorithm's precision was just 77.25%, but XGBoost's was 90.11%. The findings show that XGBoost's capacity to successfully minimize bias and variance leads in a prediction model that is more stable and dependable. Financial experts and policymakers that need accurate inflation rate estimates are advised to use XGBoost due to its

exceptional performance. It is the best option for managing intricate economic data and fluctuations connected to policy because of its sophisticated picking of features, enhancing mechanism, and against overfitting.

Declaration:

Conflict of interest:

This article has no partiality issues.

Authors Contribution:

Writing the manuscript and handling data analysis and collecting fell to author JSC. Author KVK was in charge of the planning, confirmation of data, and assessment of the text.

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Table 1: pseudocode for XGBoost algorithm.

Input: inflation rate prediction dataset
Output: improved accuracy for inflation rate prediction
1. Load the libraries needed to put the XGBoost algorithm into practice. 2. Open the application and import the dataset. 3. Separate the dataset into sets for development and assessment. 4. Create an object with the name XGBoost. 5. Train the classifier using the development set. 6. Labels for forecast test data. 7. Analyze and compare the development and assessment sets. 8. Find the percentage of precision.

Table 2: pseudocode for Decision Tree algorithm.

Input: inflation rate prediction dataset
Output: improved accuracy for inflation rate prediction
1. Load the libraries needed to put the Decision Tree algorithm into practice. 2. Open the application and import the dataset. 3. Separate the dataset into sets for development and assessment. 4. Create an object with the name dt. 5. Train the classifier using the development set. 6. Labels for forecast test data. 7. Analyze and compare the development and assessment sets and find Accuracy.

Table 3: Accuracy values of 10 iterations:

Iteration number	xg-boost accuracy	dt accuracy
1	90.07%	77.25%
2	90.21%	77.26%
3	90.21%	77.26%
4	90.15%	77.26%
5	90.01%	77.26%
6	90.02%	77.26%
7	90.01%	77.25%
8	90.16%	77.26%
9	90.21%	77.25%
10	90.10%	77.25%

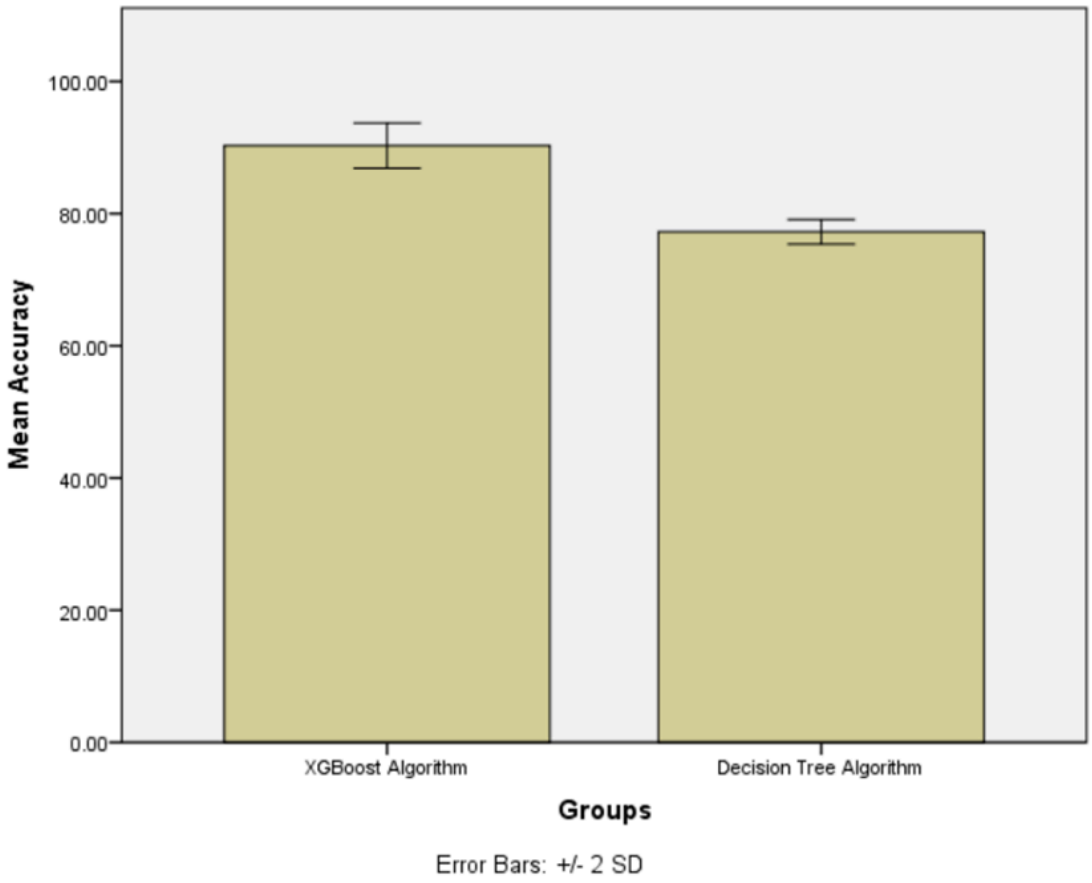
Table 4: Analysis of mean & SD

group	n	average	standard deviation	standard error mean
XGBoost	10	90.1150	0.08410	0.02659
DT	10	77.2570	0.00483	0.0153

Table 5: Independent Sample Test

independent samples test										
		levene's test for equality of variances		t-test for equality of means						
		f	sig.	t	df	sig. (2-tailed)	mean difference	std. error difference	95% confidence interval of the difference	
									lower	upper
accuracy	equal variances assumed	0.52	0.121	451.478	18	0.00	12.03	0.02665	11.97	12.08
	equal variances not assumed			451.478	9.071	0.00	12.03	0.02665	11.96	12.09

Figure 1: Extreme Gradient Boosting algorithm **Vs** Gradient boosting algorithm



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